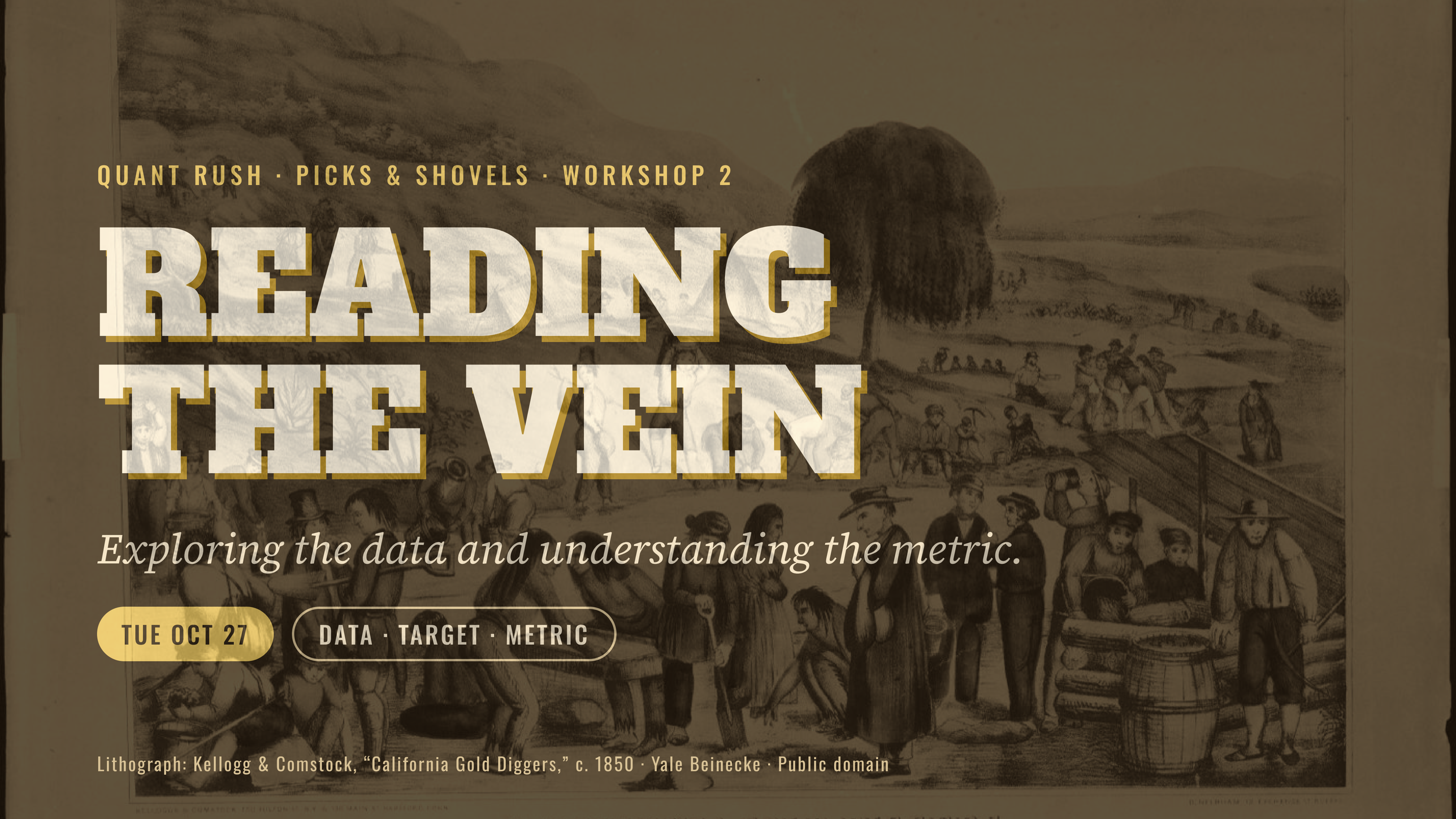
QUANT RUSH · PICKS & SHOVELS · WORKSHOP 2

READING THE VEIN

Exploring the data and understanding the metric.

TUE OCT 27

DATA · TARGET · METRIC



Lithograph: Kellogg & Comstock, “California Gold Diggers,” c. 1850 · Yale Beinecke · Public domain

TODAY

Read the ground before you dig

Last week: a first submission with one line of code. Before we build models, we learn what we are looking at.

1

THE DATA

72 columns and what each one means

2

THE TARGET

What you are really predicting

3

THE METRIC

What it rewards and what it ignores

4

THE SIGNAL

How to measure it, and how it drifts

Then: what to try on your own this week.

THE SHAPE OF THE DATA

A big table of small moments

1.18M
training rows

5,962
training eras, numbered in time order

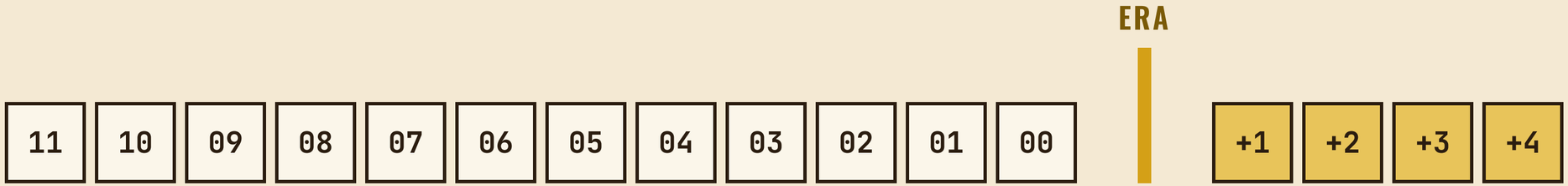
≤200
assets in each era

72
input columns per row

The test set: 328,762 rows in 1,644 eras, labeled in random order. No missing values anywhere.

ONE ROW

One asset, one era



WHAT YOU SEE: THE LAST 12 HOURS

Four channels per hour, plus the whole market that hour. **00** is the latest hour, **11** the oldest.

WHAT YOU RANK: THE NEXT 4

One number per row. Higher means you expect it to rank higher in its era.

THE COLUMNS

Four channels and the market

Columns	Each hour, the asset's…	Compared with
ret_00 … ret_11	return	the era's other assets
vol_00 … vol_11	trading volume	the era, and its own 12-hour average
range_00 … range_11	high–low price range	the era, and its own 12-hour average
taker_00 … taker_11	share of volume from aggressive buyers	the era's other assets
mkt_ret_00 … mkt_ret_11	market: the median return	same on every row of an era
mkt_disp_00 … mkt_disp_11	market: how spread out returns were	same on every row of an era

4 channels × 12 hours + 2 market columns × 12 hours = 72 inputs.

STANDARDIZED

Every column is a comparison

0

Centered in every era. +2 means far above the other assets that hour.

±5

Extremes are clipped, so one wild asset can't dominate. 2.4% of ret_00 sits at the limit.

Σ≈0

vol and range: each row's 12 values average to 0. They show change against the asset's own normal.

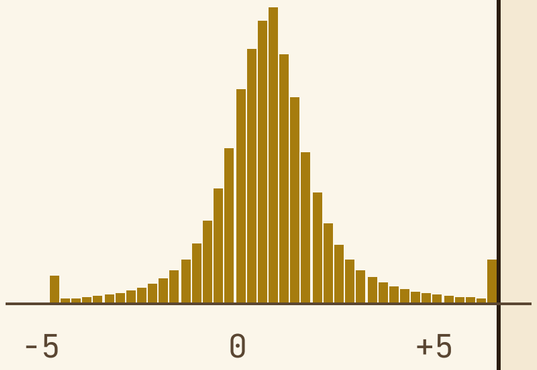
`raw return ≈ mkt_ret_k + ret_k × mkt_disp_k`

The market columns undo the standardizing, if you ever need the raw move.

DISTRIBUTIONS

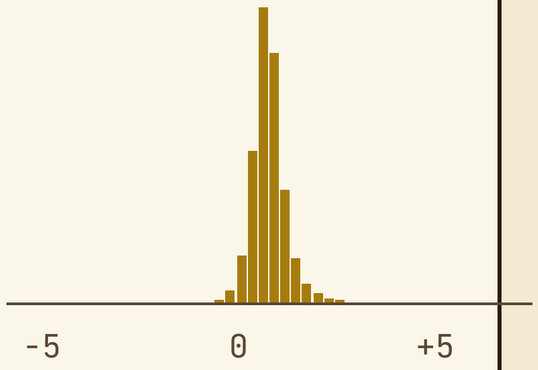
Four channels, four shapes

ret_00



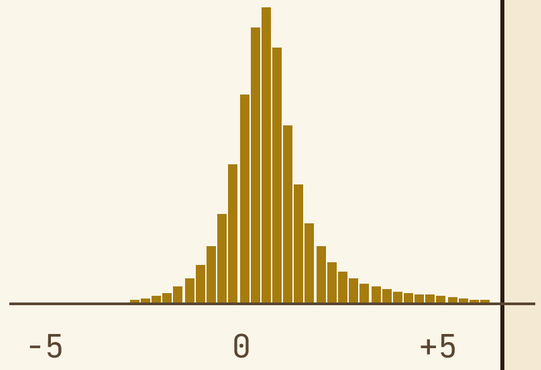
Bell-shaped with wide tails. 2.4% piles up at the ±5 clip.

vol_00



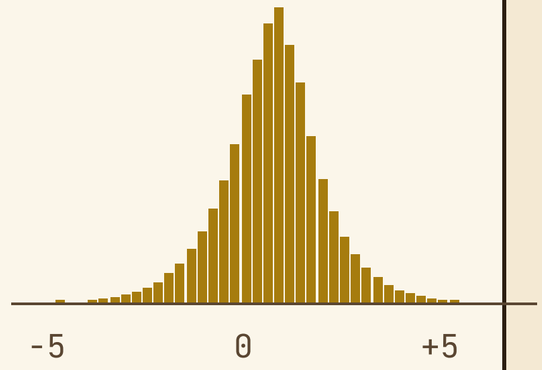
Narrow, mostly within ±1. The right tail is bursts of trading.

range_00



Skewed right: a few assets swing far more than the rest.

taker_00



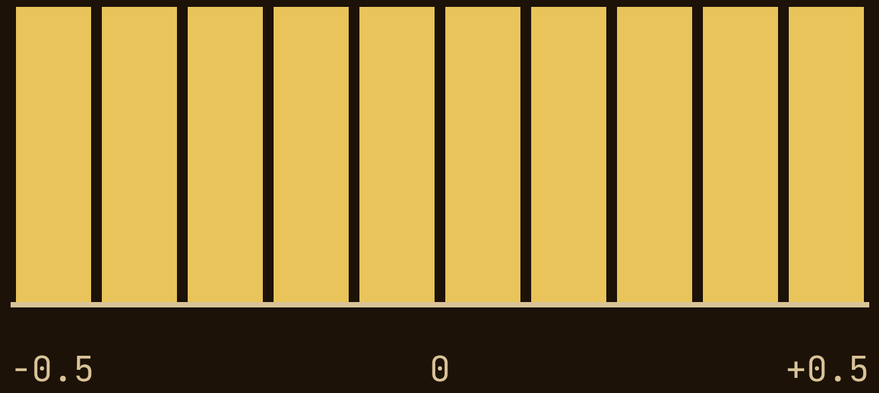
Nearly symmetric around 0. Few values reach the clip.

Tails and spikes change how a rule or a model behaves. One line turns any column into a clean rank within its era: df.groupby("era")[col].rank(pct=True)

THE TARGET

A risk-adjusted rank

1. The asset’s return over the next 4 hours…
2. …divided by its recent volatility. A 3% move means more for a calm asset than for a wild one.
3. …then ranked within the era: −0.5 is the worst asset, +0.5 the best.



The target’s histogram is flat: every era is spread evenly.

You never predict how much an asset moves. Only where it lands among the others.

THE METRIC

Spearman, era by era

1. In each era, turn your predictions into ranks: 1st, 2nd, 3rd…
2. Compare them with the true ranks. Same order scores 1, reversed −1, unrelated about 0.
3. Your score is the average over all eras.

ONE ERA, FIVE ASSETS

Asset	Your rank	True rank
A	1	2
B	2	1
C	3	3
D	4	5
E	5	4

$\rho = 0.8$ in this era

WHAT THE METRIC IGNORES

Only the order counts

The reversal rule, changed six ways, scored on the last 1,080 training eras:

Prediction	Score
-ret_00	0.0576
-ret_00 × 100	0.0576
(-ret_00) cubed	0.0576
+ a different number per era	0.0576
the same number for everyone	0.0000
sign flipped: +ret_00	-0.0576

SCALE AND SHAPE

Any change that keeps the order keeps the score.

THE WHOLE MARKET

Eras are never compared. You never predict whether everything goes up.

TIES

A constant prediction scores exactly 0.

Small scores, big swings

72%

of recent eras where the reversal rule scored above 0

0.10

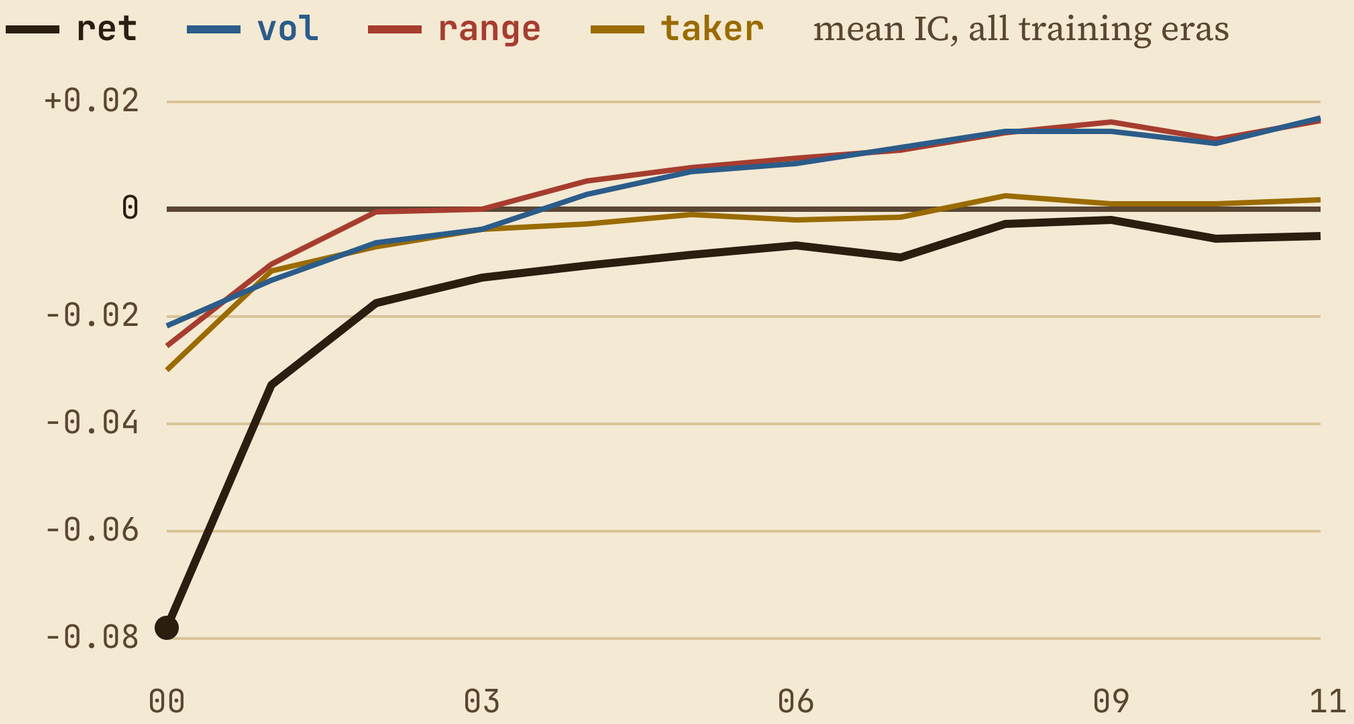
typical swing of one era's score, around an average of 0.058

±0.003

uncertainty of that average over 1,080 eras

A signal is real when it wins more often than it loses, again and again. A steady 0.05 is a good result, and scores a few thousandths apart may be equally good.

The IC: the metric, one column at a time



RET_00: -0.078

The strongest column by far, negative in 78% of eras. It fades within a few hours.

VOL AND RANGE FLIP

Negative for recent hours, positive for old ones. Why? Each row averages to 0.

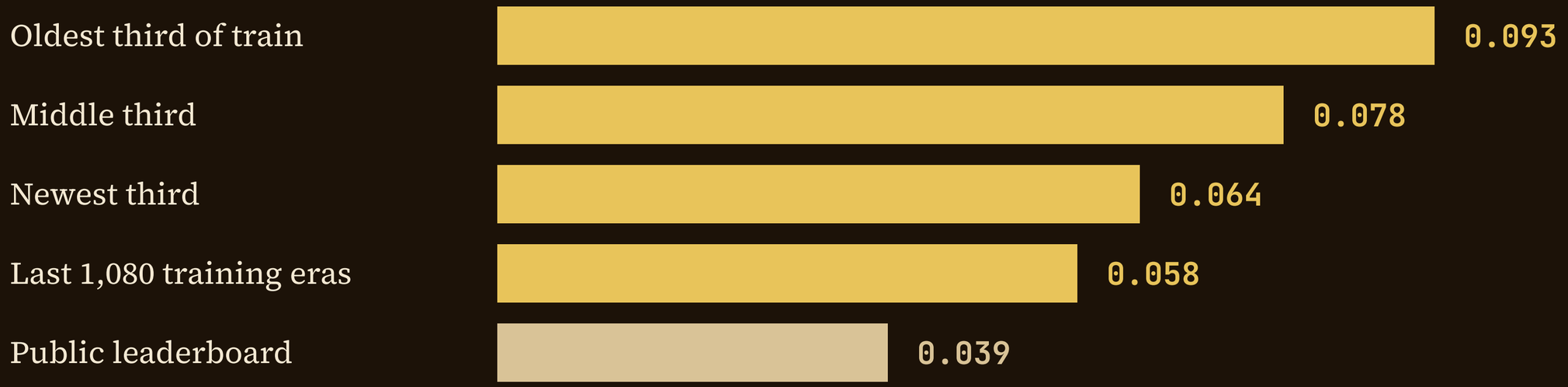
MARKET: EXACTLY 0

Same for every asset in an era, so alone it can’t rank them.

DOES IT HOLD?

Signals drift

The reversal rule’s score, from the oldest training eras to the leaderboard:



The test set comes after the training set. Check ideas on the latest training eras, not on all of train.

TRY IT THIS WEEK

Measure before you guess

```
recent = train[train["era"] > train["era"].max() - 1080]
ic = recent.groupby("era")[["ret_00", "target"]].apply(
lambda g: g["ret_00"].corr(g["target"], method="spearman"))
print(ic.mean(), (ic > 0).mean())
```

1 MEASURE

Run the lines above in your copy of the Getting Started notebook. Swap in other columns.

2 COMBINE

Mix two or three columns into one rule. Score it on recent eras with era_score.

3 SUBMIT

Only if it beats −ret_00 locally. Five submissions a day per team.

RECAP

What we found in the vein

- Every input compares an asset with the others in its era. So do the target and the metric.
- Only the order within each era counts.
- Scores are small and noisy. Consistency makes a signal real.
- Signals drift. Check ideas on the latest training eras.

Next · Tue Nov 3

REFINING THE ORE

Workshop 3: turning columns into features, and the first real models.

QUESTIONS?

Ask now, or any time on Discord: discord.gg/MUPdvSAV4M